

Recent Advances in Responsible AI

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Recall - last time

Pre-processing

- ▶ Modify the data before training the models
- ▶ Examples: reweighting, learning fair representations/feature transformation.

In-processing

- ▶ Adversarial Learning: learns a classifier and simultaneously reduces an adversary's ability to determine the protected attribute from the predictions
- ▶ Add a discrimination-aware regularization into the objective function

Post-processing

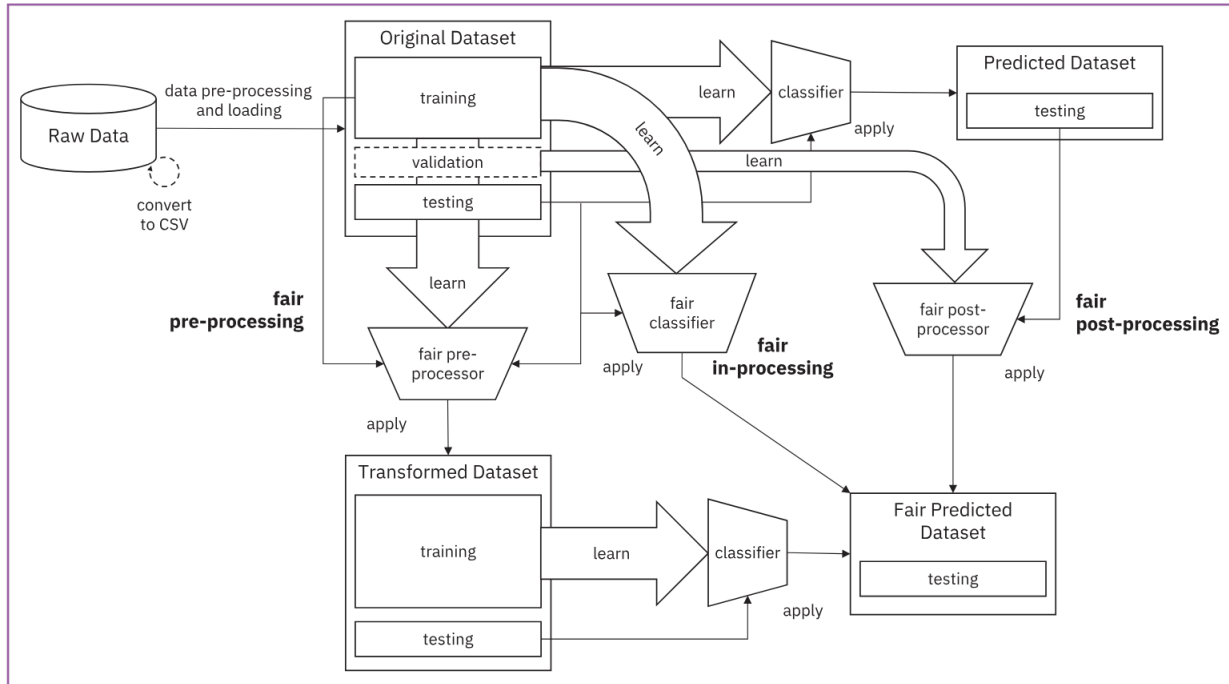
- ▶ Fair calibration of Equalized Odds
- ▶ Reject-option classification

Focus on AIF 360 (Linux Foundation & IBM)

- ▶ Documentation
<https://aif360.readthedocs.io/en/latest/index.html>
- ▶ Github
<https://github.com/Trusted-AI/AIF360>
- ▶ Demos and Tutorials:
<https://github.com/Trusted-AI/AIF360/tree/main/examples>
- ▶ ~ 10 mitigating techniques
- ▶ ~ 70 metrics (variations of Disparate Impact and Equalized Odds).
- ▶ Explainability: AIX 360 <https://aix360.res.ibm.com/>

 The perfect counterexample of what a good package documentation is.

Overview of AIF



Pre-processing Techniques in AIF

- ▶ M. Feldman et al. *Certifying and removing disparate impact*. ACM SIGKDD, 2015.
- ▶ **R. Zemel et al., *Learning Fair Representations*. ICML, 2013.**
- ▶ F. P. Calmon et al. *Optimized Pre-Processing for Discrimination Prevention*. NeurIPS, 2017.
- ▶ F. Kamiran and T. Calders, *Data Preprocessing Techniques for Classification without Discrimination*, Knowledge and Information Systems, 2012.

Learning Fair Representations (LFR)

- ▶ Paper: <https://proceedings.mlr.press/v28/zemel13.html>
- ▶ **Goal:** Learn representations that remove sensitive information while retaining predictive power.
- ▶ **Motivation:** Address bias in machine learning models by learning a transformed feature space.
- ▶ **Key concept:** Representations should be independent of sensitive attributes but still useful for prediction.
- ▶ **Fairness measure:** Both **Group Fairness** (Disparate Impact), ensuring similar distributions across groups and **Individual Fairness**.

Formalization of LFR

- ▶ Let X be the original features, S the sensitive attribute, and Y the target variable.
 - $x \in \mathbb{R}^D$ is a vector of individuals
 - $S \in \{0, 1\}$ is a binary variable indicating membership to a protected group.
 - $Y \in \{0, 1\}$ is the binary random variable representing the classification decision
- ▶ X_0 denotes the training set of individuals.
- ▶ $.^+$ and $.^-$ indicates individuals with $S = 1$ and $S = 0$, respectively.
- ▶ Z is a multinomial random variables of K events.
 - v_k is the vector of the k -th prototype.

Objective of LFR

The probability that a random element from X^+ maps to a particular prototype is equal to the probability that a random element from X^- maps to the same prototype.

$$\mathbb{P}(Z = k | x^+ \in X^+) = \mathbb{P}(Z = k | x^- \in X^-).$$

Representation: $g : X, Y \rightarrow Z$ via prototypes, defined by parameterized mixture model that stochastically maps data to prototypes.

Prediction: $f : Z \rightarrow Y$ Parameterized mixture model that stochastically maps prototypes to labels

Representation: $X \rightarrow Z$

- ▶ Discriminative clustering model to map from X to Z .
- ▶ The probability of $Z = k$ given x follows a soft assignment based on a distance function $d(x, v_k)$.

$$\mathbb{P}(Z = k | x \in X) = \frac{\exp(-d(x, v_k))}{\sum_{k'=1}^K \exp(-d(x, v_{k'}))},$$

where $d(., .)$ is the euclidean distance.

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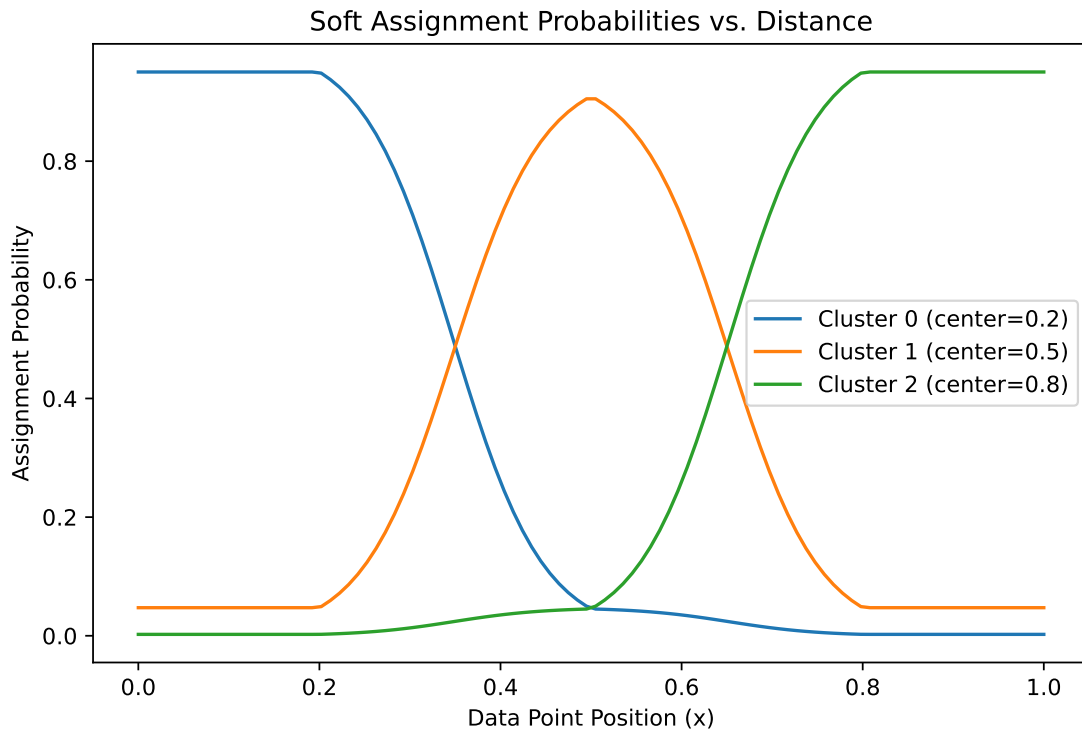
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Remark: a smaller distance $d(x, v_k)$ leads to a higher probability of assigning x to prototype k .

Illustration of $X \rightarrow Z$



Redefinition of the objective

The objective of LFR can be defined by these three goals:

1. the mapping from X_0 to Z satisfies statistical parity;
2. the mapping to Z -space retains information in X (except for membership in the protected set);
3. the induced mapping from X to Y through Z is close to the direct mapping from X to Y .

The loss function reflects these three objectives

$$L = A_z.Lz + A_x.Lx + A_y.Ly,$$

where A_z , A_x and A_y are hyper-parameters allowing to control the utility-fairness tradeoff.

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This equation

$$\mathbb{P}(Z = k | x^+ \in X^+) = \mathbb{P}(Z = k | x^- \in X^-).$$

can be expressed as

$$M_k^+ = M_k^-, \quad \forall k,$$

where $M_k^+ = \mathbb{E}_{x \in X^+} \mathbb{P}(Z = k | x) = \frac{1}{|X_0^+|} \sum_{n \in X_0^+} \mathbb{P}(Z = k | x_n)$.

- ▶ M_k^+ represents the expected (average) probability that a sample from the set X^+ is assigned to the cluster k .
- ▶ If M_k^+ is large, it means that cluster k is mostly composed of points from X^+ .

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$$L_z = \sum_{k=1}^K |M_k^+ - M_k^-|$$

L_x : the mapping to Z retains information on X

- Quantify the reconstruction error as

$$L_x = \frac{1}{n} \sum_{n=1}^N (\hat{x}_n - x_n)^2,$$

where $\hat{x}_n$ are the reconstruction of x_n from Z :

$$\hat{x}_n = \sum_{k=1}^K \mathbb{P}(Z = k | x_n) v_k.$$

- We take a weighted combination of all cluster centers to approximate x_n .
- The goal of minimizing L_x is to ensure that the learned clusters v_k capture enough information about X so that we can still reconstruct it.

L_y : the mapping allow to predict Y

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$$L_y = \sum_{n=1}^N -y_n \log(\hat{y}) - (1 - y_n) \log(1 - \hat{y}_n),$$

where

$$\hat{y}_n = \sum_{k=1}^K \mathbb{P}(Z = k | x_n) w_k$$

- w_k 's are constrained to be in $[0, 1]$.
- w_k is the prototype prediction for cluster k , i.e., the probability that a point in cluster k belongs to class 1.
- $\hat{y}_n$ is computed as a weighted sum of prototype prediction, using the soft assignment probabilities as weights.

Numerical Example

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$$\hat{y}_n = M_{n,1}w_1 + M_{n,2}w_2 + M_{n,3}w_3$$

$$\hat{y}_n = (0.6 \times 0.8) + (0.3 \times 0.5) + (0.1 \times 0.2)$$

$$\hat{y}_n = 0.48 + 0.15 + 0.02 = 0.65$$

Thus, the predicted probability of class 1 for x_n is **0.65**.

Summing-up LFR

The goal is to minimize

$$L = Az.Lz + Ax.Lx + Ay.Ly,$$

- ▶ Parameters: $\mathbf{v}_k, w_k, \forall k$;
- ▶ Hyper-parameters: A_z, A_x and A_y .

Additional variant: different input features can have different levels of impact.

$$d(x_n, v_k, \alpha) = \sum_{i=1}^D \alpha_i (x_{ni} - v_{ki})^2$$

- ▶ Optimization is done with L-BFGS: a gradient-based method suitable for convex and smooth non-convex functions.

In-processing Techniques in AIF

- ▶ B. H. Zhang et al., ***Mitigating Unwanted Biases with Adversarial Learning***. AAAI 2018
- ▶ M. Kearns et al., *Preventing Fairness Gerrymandering: Auditing and Learning for Subgroup Fairness*. ICML 2018.
- ▶ L. E. Celis et al., *Classification with Fairness Constraints: A Meta-Algorithm with Provable Guarantees*, 2018.
- ▶ T. Kamishima et al., ***Fairness-Aware Classifier with Prejudice Remover Regularizer***. ECML-PKDD 2012.
- ▶ A. Agarwal et al., *A Reductions Approach to Fair Classification*. ICML, 2018.

Adversarial Learning for Fair Representation

- ▶ A classifier learns to make predictions, while an adversary tries to predict the sensitive attribute.
- ▶ If the adversary cannot predict the sensitive attribute, it means that the learned representation does not contain bias.
- ▶ The predictor is trained to maximize accuracy, while the adversary is trained to minimize the amount of bias

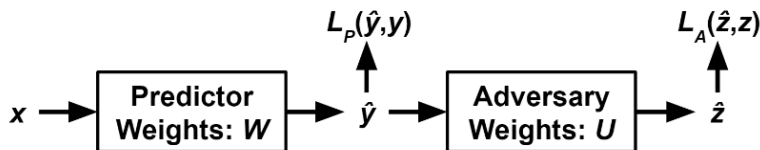


Figure 1: The architecture of the adversarial network.

Objective Function

The total objective consists of two competing terms:

$$\min_W \max_U \mathcal{L}_P(f_W(x), y) - \lambda \mathcal{L}_A(g_U(f_W(x)), s)$$

where:

- ▶ $\mathcal{L}_P$ is the prediction loss (e.g., cross-entropy loss for classification).
- ▶ $\mathcal{L}_A$ is the adversary's loss (trying to predict the sensitive attribute s).
- ▶ $f_W(x)$ is the output layer of the network.
- ▶ $g_U(f_W(x))$ is the adversary's prediction.
- ▶ λ controls the trade-off between fairness and accuracy.

Summing up Adversarial Learning for Fairness

- ▶ Optimisation is done through an alternating Optimization Procedure.
- ▶ Depending on the fairness metric optimized, the input of the adversary might change.
 - e.g. for equalized odds, the input is the output layer $\hat{y}$ and the true label y .
- ▶ Comes with theoretical guarantees **under optimal adversary performance**.
- ▶ Can be applied to various models (e.g., deep learning, CNNs, NLP tasks).

Post-processing Techniques in AIF

- ▶ **G. Pleiss et al., *On Fairness and Calibration*. NeurIPS 2017.**
- ▶ F. Kamiran et al., *Decision Theory for Discrimination-Aware Classification*, ICDM, 2012.
- ▶ S. Cem Geyik et al., *Fairness-Aware Ranking in Search & Recommendation Systems with Application to LinkedIn Talent Search*. KDD 2019.

What is Calibration in ML?

Definition: calibration refers to the alignment between predicted probabilities and actual observed frequencies.

Goal: if a model assigns a probability of 0.7 to an event, we expect that event to occur approximatively 70% of the time.

e.g. a model predicting 80% probability of a disease should be correct 80% of the time.

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Why is calibration important?

- ▶ **Overconfident models:** Lead to unnecessary interventions or high risk-taking.
- ▶ **Underconfident models:** Reduce trust in predictions, leading to missed opportunities.

Calibration: a numerical example (1)

Borrower	Model 1	Model 2	True Y
A	0.9	0.99	1
B	0.8	0.95	1
C	0.7	0.90	1
D	0.6	0.85	1
E	0.5	0.80	0
F	0.4	0.80	0
G	0.3	0.70	1
H	0.2	0.60	0
I	0.1	0.50	0
J	0.05	0.40	0

Question: which model is well-calibrated?

Calibration: a numerical example (2)

Let's make some bins

Bin Range	M1 Avg. Pred.	M1 Obs. Freq	M2 Avg. Pred.	M2 Obs Freq
[0.0 - 0.2]	0.12	0	—	—
]0.2 - 0.4]	0.35	0.5	0.4	0
]0.4 - 0.6]	0.55	0.5	0.55	0
]0.6 - 0.8]	0.75	1	0.77	$\frac{1}{3}$
]0.8 - 1.0]	0.9	1	0.92	1

We can also compute the expected calibration error following this principle (if we have time).

Fairness and Calibration

- ▶ Well-calibrated means that among all instances where the model predicts $\hat{P} = p$, the actual proportion of $Y = 1$ should be approximately p .

$$P(Y = 1 | \hat{P} = p) \simeq p$$

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- ▶ Maintaining cost parity and calibration is desirable yet often difficult in practice (impossibility results)
- ▶ Fair calibration refers to the idea that a model should be well-calibrated across different demographic group

$$P(Y = 1 | \hat{P} = p, A = a) \simeq p, \forall a$$

Final Takeaways: Fairness in Machine Learning

Different Approaches, Different Trade-offs

- ▶ **Pre-processing:** Adjusts data before training → Ensures fairness at the input level but may distort information.
- ▶ **In-processing:** Modifies the learning process → More flexible but often computationally expensive.
- ▶ **Post-processing:** Adjusts predictions after training → Easiest to apply but constrained by the learned model.

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No One-Size-Fits-All Solution

- ▶ Each method has strengths and limitations depending on the use case, constraints, and ethical considerations.

Final Takeaways: Fairness in Machine Learning

Fairness is Still an Open Research Area

- ▶ Defining fairness formally remains challenging.
- ▶ Trade-offs between fairness and accuracy need deeper understanding.
- ▶ Ongoing work explores long-term impacts and fairness beyond individual predictions (e.g., fairness in dynamic settings, intersectionality).

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Beyond Static Models: Future Directions

- ▶ Fairness-aware calibration, robustness to distribution shifts, and domain-specific fairness notions are emerging topics.
- ▶ Increasing interest in **causal fairness** and fairness-aware generative models.